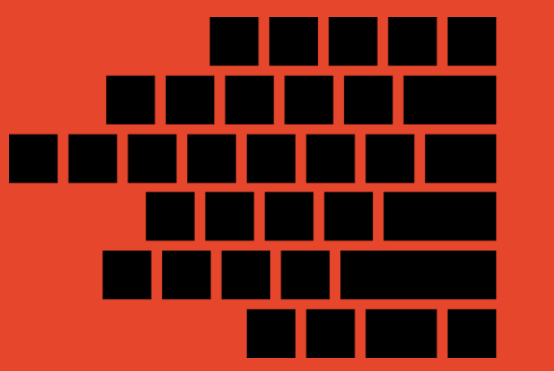




Project Description (e.g., Background and Aims/Objectives)

An interactive VST plugin that acts as your mentor, not your ghost producer.



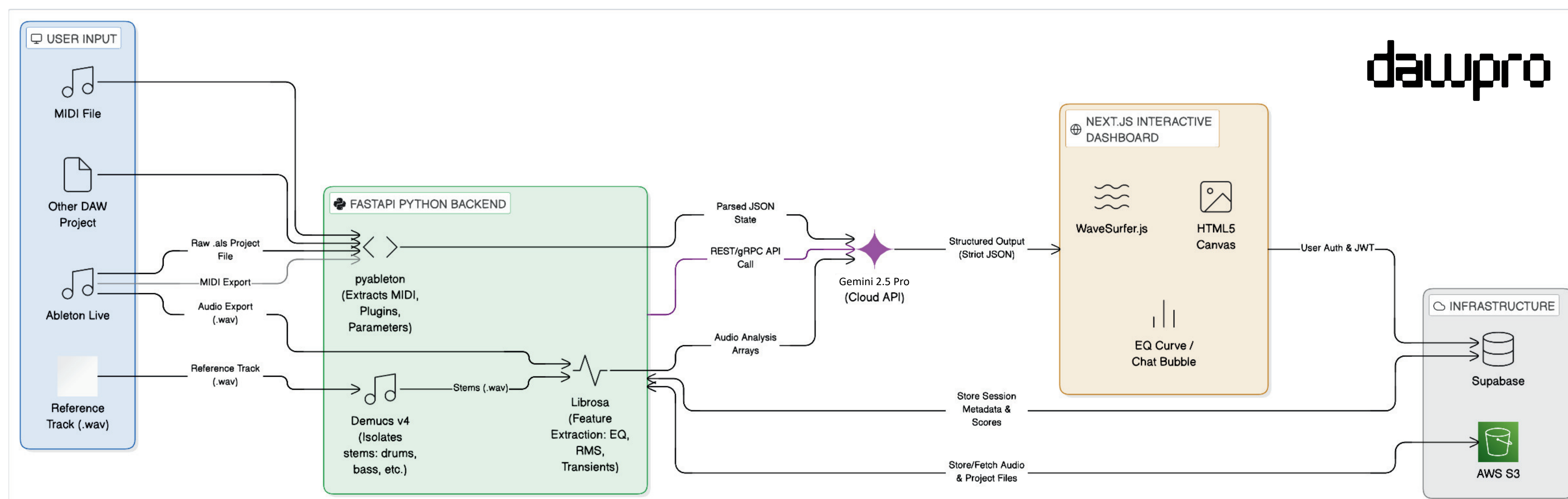
The democratisation of digital audio workstations (DAWs) had led to a massive surge in self-taught music producers. However, the learning curve for music theory, sound design and audio engineering remains a steep learning curve. **DAWpro** aims to bridge the gap between amateur experimentation and professional mentorship by being an interactive co-producer providing contextual technical and actionable feedback to accelerate skill acquisition.

Project Solution & Results

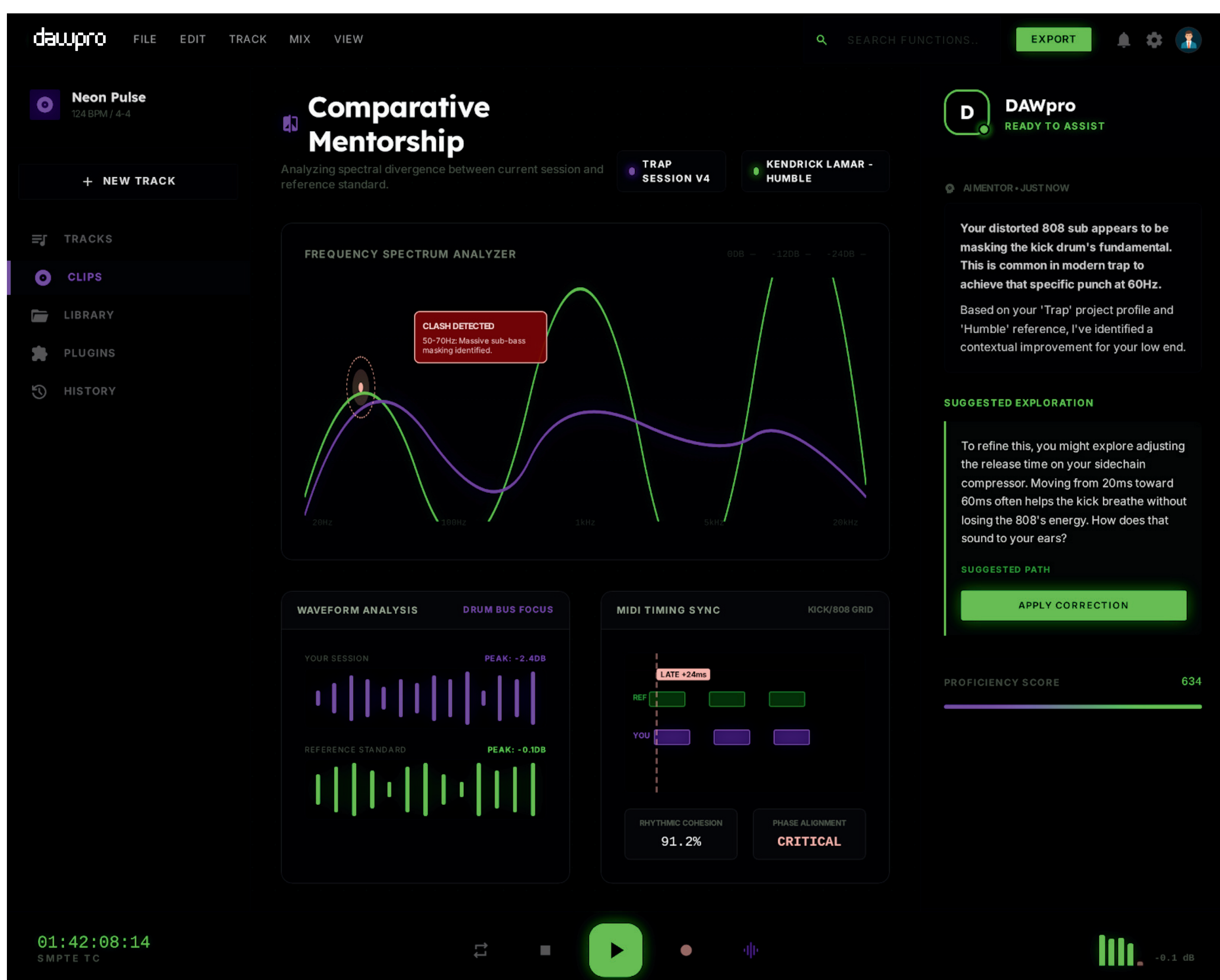
DAWpro is designed as a learning tool first, with skill acquisition for self-taught music producers being a top priority. **DAWpro** is built as a dual-environment ecosystem: A lightweight **C++/JUCE** VST plugin paired with **Next.js** and **FastAPI** web application. The VST acts as a bridge, silently capturing real-time audio and metadata directly from the DAW. Core features include the ability for users to upload reference tracks, set specific sonic intentions, and visualize deep comparative analytics through an interactive web dashboard.

Technical Architecture & Data Pipeline

To deliver real-time, context-aware mentorship, DAWpro relies on a robust, multi-layered architecture:



- **The Bridge (C++/JUCE VST):** Sits natively on the master bus, silently capturing high-fidelity audio streams and extracting vital DAW metadata without disrupting the producer's workflow.
- **The Engine (FastAPI & Python):** Handles heavy audio processing. It utilizes **Demucs v4** for high-quality stem separation of reference tracks, and **Librosa** to extract detailed audio features (RMS levels, transient peaks, frequency spectrums).
- **The Brain (Gemini 2.5 Pro & LangChain):** The parsed audio arrays and user intentions are fed into the LLM. Leveraging its massive context window, the AI analyzes the data to output structured JSON diagnosing mix issues and suggesting technical solutions.
- **The Dashboard (Next.js):** A responsive web interface that translates the AI's JSON output into interactive visual overlays (like comparative EQ curves and timing grids), built with **Tailwind CSS** and **WaveSurfer.js**.



DAWpro in Action: Contextual Mentorship

Rather than offering generic mixing advice, **DAWpro** tailors its feedback to the exact genre and acoustic profile of the project.

For instance, if a producer is working on a modern trap beat or a trance track in Ableton, they can input their sonic intention. DAWpro's comparative analytics will cross-reference the extracted stems and identify specific clashes. The dashboard might visually highlight certain issues.

It is important to note that music is subjective, hence we would never give a user specific instructions unless instructed to do so. Instead, **DAWpro** will help guide users towards the right direction to help facilitate sonic exploration.